ComPy inversion overhaul, November 2026

Audit and rebuild of the shear-velocity inversion of ComPy, against the primary literature and against tests that do not use any earlier ComPy result. Branch `inversion-overhaul`, proposed as ComPy 3.0.0. Working notes: NOTES.md.

Every statement below is marked **[test]** (verified by a script in `scripts/`, output in `~/Desktop/ComPy/14_inversion_overhaul/`), **[source]** (taken from a publication that was read) or **[inferred]**.

1. Summary

What was wrong in the inversion of ComPy 2.3 to 2.8

1. **The error model.** Each frequency bin carried an independent 1 % error. The real scatter of a stacked compliance curve is 0.2 to 0.4 % and strongly correlated between bins; what is uncertain is the level of the curve (pressure gain) and its slope (pressure response at low frequency). The old model was five times too loose on the shape of the curve, which explains chi-square/N near 0.4, and too strict on its slope, which is what carries the information about the deep crust. [test]
2. **The prior.** Layers had a flat prior with a penalty on second differences. In the layers the data do not see, that penalty continues the shallow gradient and drives Vs to its bounds. This is the "deep layers press against their limits" of the earlier reports. [test]
3. **Vp and density.** Compliance measures stiffness, not Vs. One free Vp/Vs for all crustal layers settled near 2.1, which gives lower-crust Vp above 8.5 km/s and, with the density rule tied to Vp, a density of 3.3 g/cc. Both lower Vs: by about 10 % at 2 to 5 km. [test]
4. **Sediment.** A soft layer much thinner than the wavelength adds compliance through its P modulus, `k h / (rho Vp^2)`, whatever its Vs. The model fixed sediment density at 2.0 g/cc (soft sediment is nearer 1.6) and its Vp, and limited thickness to 1 km. Sediment thickness, and through it deep Vs, inherited those choices. [test]
5. **Fixed Moho** from a 1-degree model, and **uniform gain bounds** of 5 %, where two independent calibrations of the same gauges differ by more than that in one case out of six. [test]
6. **Band.** At shelf stations the estimate collapses above the frequency where the pressure signal still reaches the seafloor; the coherence gate does not detect it, and the saved curve was also cut at 40 mHz. Shelf station 7D.J09B never fitted for these reasons. [test]

What was right. The forward model (agreement with an independent 50-digit solution to 4e-7), the gravity corrections of the data (re-derived; the 2025 paper states their sign and size wrongly, the code is right), the sampler (acceptance rules read and tested; a ComPy 2.8 run repeats draw for draw), and the H1 estimator. [test]

What changed. A new default inversion, `compy.inversion.auto`, with errors measured from each station's own windows, a proper prior about a reference model, free Moho depth, Vp and density uncertain, sediment with its P modulus, a Gaussian gain prior, automatic band limits, sampling that repeats until it converges, and a result that marks the depths the data resolve. One command: `compy-invert`. Method: ../INVERSION.md.

Is it calibrated? On synthetic data drawn from the prior, on four station types, the 50 % and 90 % intervals contain the truth at those rates (section 5). When the true Earth is one the model cannot represent, intervals stay honest below 700 m and Vs in the top 300 m follows the reference, not the data; the result flags those depths as unresolved. [test]

What the data resolve. At a deep-ocean station: the thickness of soft sediment, and broad averages of Vs between about 0.7 and 6 km. Not the top few hundred metres of basement, not thin layers, and nothing about Vp/Vs or density. [test]

How much the Vs models moved (archive rerun, 155 tier-1 stations that are not "poor"; ComPy 3 against ComPy 2.8, median and 16 to 84 % range):

Depth	Change of median Vs	Stations with this depth resolved
0.5 km	-4 % (-18 to +19)	27 %
1 km	+4 % (-18 to +18)	43 %

Depth	Change of median Vs	Stations with this depth resolved
2 km	+6 % (-7 to +15)	82 %
3 km	+7 % (-5 to +17)	86 %
5 km	+11 % (-1 to +20)	56 %

Soft-sediment thickness is 0.76 times the 2.8 value (median). At the eight stations of the 2025 paper the change is larger: Vs at 2 to 3 km rises by 4 to 16 % at six of them and falls by 9 % at RR29; at RR38, RR40, RR50 and RR52 the new medians at 2 to 5 km are 4.1 to 4.7 km/s (section 7.2). These numbers compare two ComPy versions; neither is a reference.

Confidence. High for the diagnosis of the error model, the prior and the forward model: each rests on a test. Moderate for absolute Vs below 2 km: it depends on the Vp/Vs and density priors and on the slope allowance, whose sizes are judgements stated in section 4 and tested in sections 5 and 8. Lower for Vs between 0.5 and 3 km where sediment is thick: there it depends on where the band is cut (section 8). The independent sediment map agrees with the soft-sediment thickness where sediment is thin (rank correlation 0.66 over 92 stations, section 7.3).

What failed or stays open is in section 9; **decisions for the author and W. Crawford** in section 10.

2. Literature

Source	Read	What was taken
Crawford (1994), PhD thesis, UCSD	ch. 2 and 6 (ch. 3 and 4 are placeholders in the PDF)	Half-space compliance, eq. 2.19; sensitivity ratios, eqs 2.20 to 2.22; Vp fixed from refraction; start Vs = Vp/1.87; density of Carlson & Raskin (1984), eq. 6.1; Occam inversion; sensitivity near lambda/6
Crawford (2004), GJI 157	in full	Definition, eq. 1; `eta = (lambda + 2 mu)/(2 mu (lambda + mu))`, eq. 5; "density has almost no effect"; depth between lambda/2pi and lambda/4; error formula, eq. 8; minimum-structure and Bayesian inversions; full equations of motion always solved
Zha (2014), PhD thesis, Columbia (holds Zha et al. 2014 and Zha & Webb 2016)	ch. 2 and 4	Gravity correction as Crawford et al. (1998) eq. 6 without an errant 2 pi, sign of its effect; DRAM; sediment 0 to 2000 m with fixed Vs(z) (Ruan) and Vp(z) (Hamilton), density 2.0; crust layers 100 m doubling; Vp/Vs 1.85; density 1.85 + 0.165 Vp; mantle 3.8 to 4.2 km/s; Moho from refraction
Hulme et al. (2003, 2005); Crawford et al. (2002); Crawford (2005)	sections	2-D effects, as a limit
Mosher et al. (2021), GJI 227	sections	Another parameterization (Bernstein polynomials, constant Vp and density)
Janiszewski et al. (2019), GJI 217	one statement	Sediment density 2.0 "following Ruan et al. (2014)"
Tsekhmistrenko et al. (2026), Seismica 5(2)	sections	Instruments of network 8J
Aminian et al. (2025), GJI 242; Aminian (2024) thesis	earlier audits	The method under test

Not obtained (publisher or host refused automated access, or closed): Crawford et al. (1991, 1998); Ruan et al. (2014); Bell et al. (2015); Doran & Laske (2019); Zheng et al. (2024); Brocher (2005), Hamilton (1976, 1979), Carlson & Raskin (1984), Christensen & Shaw (1970) in the original. Consequences: the Ruan sediment relation is not used; Brocher's relations are taken from a published implementation (R package hydrogeo) and marked so in the code; the 1998 gravity formula is taken from Zha's quotation and re-derived (section 3.1). Please check these against the originals.

3. Physics checks

3.1 Gravity terms of the data

With z up, a pressure p pushes the seafloor down, `u = -xi p`. A seismometer reads specific force, `a = d2u/dt2 - dg`. A wave crest attracts the mass upward, `dg = a_w = (2 pi G / g) cosh(kH) exp(-kH) p`; moving down by |u| raises gravity by the free-air gradient, `dg = (2g/R) u`. Hence `a/p = (omega^2 + 2g/R) xi - a_w/p`, and

`eta = k (omega^2 |Z/P| + a_w/p) / (omega^2 + 2g/R),`

which is what `compy.compliance.gravity_corrected_compliance` computes. The correction raises compliance, as Zha et al. (2014) say. At YV.RR52 it is 18 % of compliance at 7.3 mHz and 6 % at 17 mHz. The 2025 paper says the attraction "can increase the measured compliance by up to 5 per cent": sign and size are wrong in the text, not in the code. The attraction of the displaced seafloor itself is a fifth of the free-air term, which is 0.3 % at 5 mHz: negligible. [inferred from first principles; formula of a_w from source]

3.2 Forward model

A global-matrix solver written for this audit (all layer amplitudes in one linear system, 50-digit arithmetic) agrees with `dram.Forward` to 3.6e-7 and with `impedance` to 2.8e-8 over 60 random models, half of them with slow sediment, under 150 to 5000 m of water. A gradient split into n sublayers converges as 1/n^2 (16 sublayers: 0.3 %). [test: `e01_forward_checks.py`]

3.3 What compliance is sensitive to

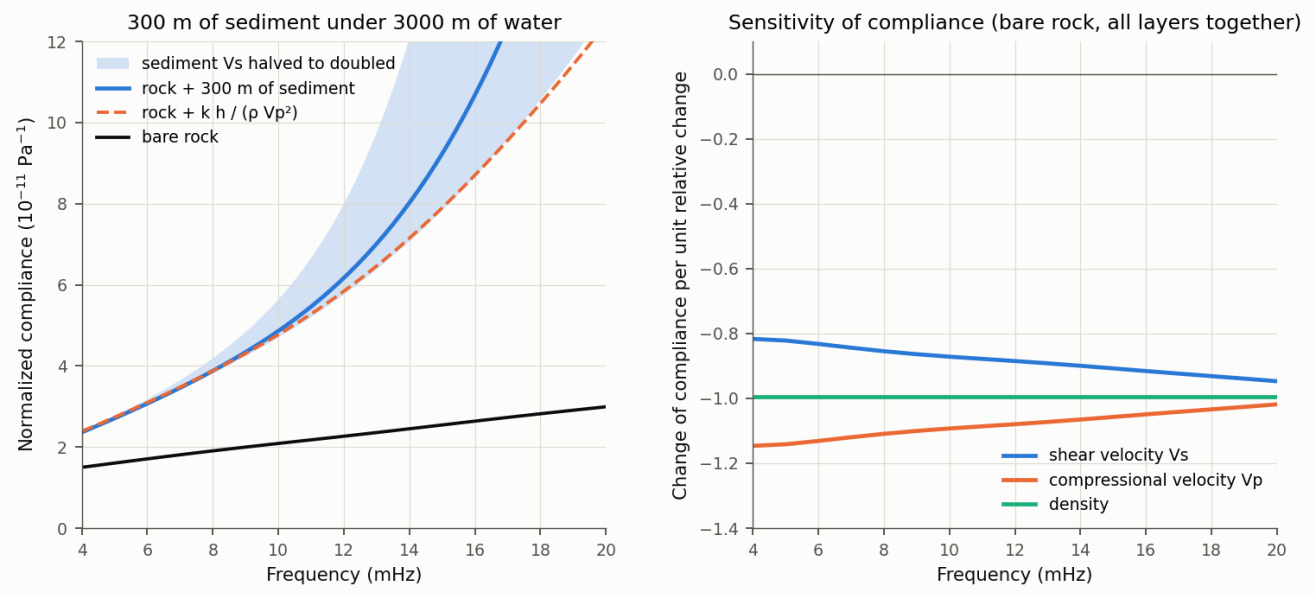


Figure 1. Left: compliance of an oceanic crust under 3000 m of water without sediment (black) and with 300 m of sediment (blue); the dashed line adds only the uniaxial term k h/(rho Vp^2) to the bare rock; the band is the range when the sediment Vs is halved or doubled. Below 10 mHz the sediment acts through its P modulus alone. Right: change of compliance for a 1 % change of Vs, Vp or density in all layers. Read it as: compliance is about as sensitive to Vp as to Vs.

- Half-space: `eta = (1 - nu)/mu`. Between Vp/Vs 1.73 and 2.4 the same compliance gives Vs 11 % apart. [source: Crawford 2004 eq. 5; arithmetic]
- Summed over a normal crust, `d ln eta/d ln Vs` is -0.80 to -0.96, `d ln eta/d ln Vp` -1.02 to -1.17, `d ln eta/d ln rho` -0.99. The published "2 to 9 times more sensitive to Vs" is per m/s. [test]
- Thin soft layer: 300 m of sediment adds 58 % (4 mHz) to 133 % (10 mHz) to bare-rock compliance; the added part equals `k h/(rho Vp^2)` to 3 % up to 10 mHz; halving the sediment Vs changes compliance by 1 % at 4 mHz and 3 % at 6 mHz. [test, both forward codes and the 50-digit solver]
- Attenuation: seismic velocities at 5 to 10 Hz exceed those at 10 mHz by `ln(700)/(pi Q)`: 0.4 % for Q = 500, 4 % for Q = 50, 10 % for Q = 20. Compliance Vs of sediment and of cracked upper crust should be read a few per cent below refraction Vs for this reason alone. [inferred: standard constant-Q dispersion, not re-read]

4. Errors of a compliance curve

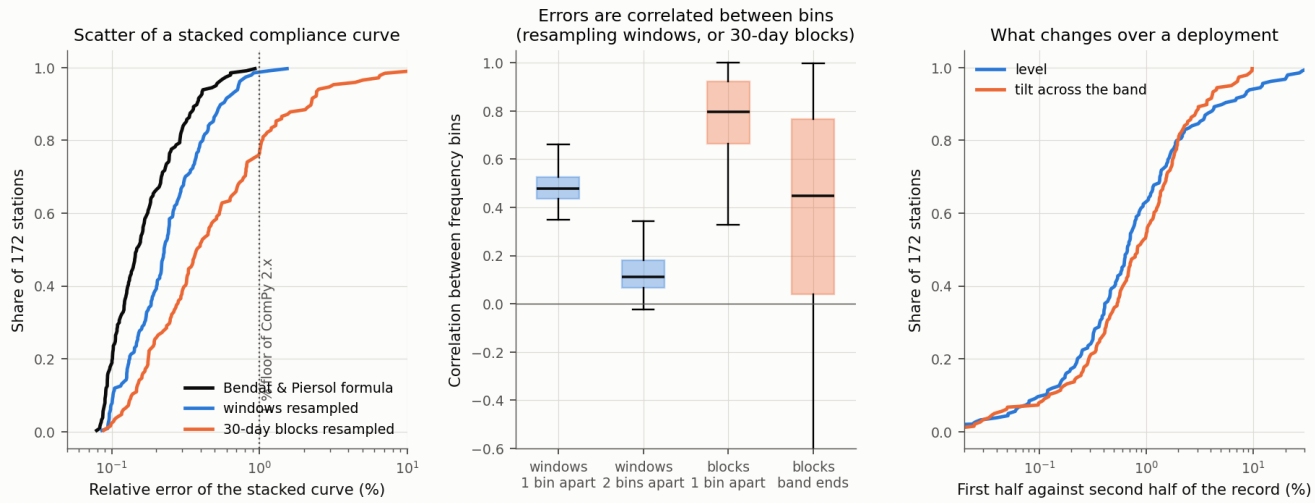


Figure 2. 172 stations of the archive. Left: share of stations whose stacked curve has at most the given relative error, by three measures; the dotted line is the 1 % floor of ComPy 2.x. Middle: correlation of the errors between frequency bins. Right: change of the level and of the tilt of the curve between the two halves of the record.

Quantity (172 stations)	Median [10 % to 90 %]
Bendat & Piersol error	0.15 % [0.09 to 0.38]
Bootstrap over windows	0.23 % [0.10 to 0.55]; 1.36 times Bendat & Piersol
Correlation of neighbouring bins; two apart	0.48; 0.11
Bootstrap over 30-day blocks	0.38 % [0.14 to 2.2]
Share of the 30-day covariance that is a level / a tilt	59 % / 11 %
First half against second half: level; tilt	0.67 %; 0.83 %
H2/H1 of the stack	1.09
Compliance of the loudest fifth of windows over the quietest fifth	0.985 [0.88 to 1.003]

[test: `e02_error_model.py`; the restack equals the archive to 1.5e-14]

- **Estimator.** Noise on the pressure channel would make H1 rise with the pressure level. It falls slightly. So coherence is limited by noise on the vertical, H1 is the right estimator, and H2 (9 % higher) is not a useful bracket. The small fall is in the direction of the selection bias of the coherence gate found in the 2.9 audit. [test]
- **Gain.** The Rayleigh-wave gain and the microseism gain of 73 gauges differ by 2.7 % (robust scatter; 2.0 % for the 36 microseism pairs), with a 16 to 84 % range of -5.4 to +2.6 % and outliers beyond 25 %. A Gaussian prior of 5 % is supported; a hard +-5 % is not. [test]
- **Slope.** The tide-based correction of the pressure-gauge leak corner (2.9 audit) changes compliance by -3 % (median) at the bottom of the band, -6 % at YV.RR52. The allowance for a slope error is 2 % where that correction is applied, more where the tide and the phase estimate of it disagree, 4 % where no correction exists. This number is a judgement; section 5 shows what happens when it is exceeded. [inferred]

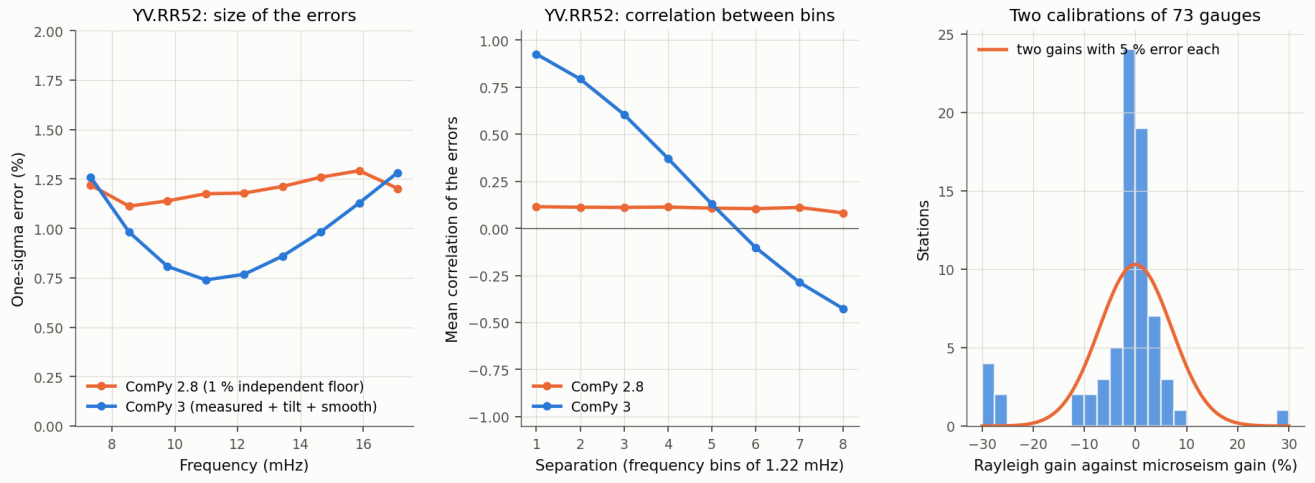


Figure 3. Left and middle: the errors given to the inversion at YV.RR52 by ComPy 2.8 and by ComPy 3: their size, and their correlation between bins. Right: difference between two independent pressure calibrations; the curve is what two gains with 5 % error each would give.

5. Calibration of the new inversion

Synthetic data sets, 144 per case: a true model is drawn, its compliance computed, noise added with the station's covariance, and the result inverted (`e03_sbc.py`). For 144 cases a correct 50 % interval contains the truth in 42 to 58 % of them and a 90 % interval in 85 to 94 %.

Case	Parameters within both ranges	Vs at 8 depths, 50 %	Vs, 90 %	Median bias of Vs
Truth from the prior, YV.RR52 (deep, 9 data)	15 of 16	0.39 to 0.60	0.83 to 0.92	-1.1 to +0.4 %
Truth from the prior, 7B.NN04 (abyssal, 8 data)	15 of 16	0.45 to 0.54	0.85 to 0.91	-2.4 to +0.7 %
Truth from the prior, 7K.AN3 (slope, thick sediment, 13 data)	15 of 17	0.46 to 0.55	0.88 to 0.94	-1.4 to +1.4 %
Truth from the prior, 7D.J09B (shelf, 23 data)	17 of 18	0.42 to 0.55	0.90 to 0.94	-2.9 to +0.9 %
Smooth-gradient Earth the layers cannot represent, RR52	n/a	0.49 to 0.78 from 700 m down; 0.40 to 0.53 above 300 m	0.97 to 1.00 from 700 m down; 0.66 to 0.94 above 300 m	-3.0 to +1.9 % from 700 m down; +16 to +25 % above 300 m
The same, NN04	n/a	0.51 to 0.79; 0.35 to 0.50	0.91 to 1.00; 0.72 to 0.95	-5.3 to +8.3 %; +16 to +26 %
Unmodelled slope error of 4 %, RR52	15 of 16	0.37 to 0.59	0.82 to 0.92	-2.8 % at 50 m to +7.0 % at 7 km
Gain error of 10 % (twice the prior), RR52	13 of 16 (the gain itself: 0.17 and 0.56)	0.33 to 0.50	0.78 to 0.88	-1.4 to +0.8 %

[test; full numbers in `e03_sbc/summary.json`]

Reading:

- With the truth drawn from the prior, coverage is nominal for every parameter type on all four station types. Five of 67 parameter rows fall just outside one of the two ranges (coverage 0.83 to 0.95 against 0.85 to 0.94, or 0.59 to 0.60 against 0.58), which is what chance gives for this many rows.
- With an Earth of smooth gradients, a depth-dependent Vp/Vs between 1.76 and 2.5 and another density rule, intervals below 700 m are conservative and the median error of Vs is 2 to 9 %. In the top 300 m the truth has a slow layer 2A, the reference does not, and the data cannot tell: Vs there is the reference. These are the depths the result marks as unresolved.

- A slope error twice the allowance biases Vs below 3 km by 2 to 7 %. This is the main systematic risk of deep Vs.

6. The earlier sampler still repeats

Variant `final` of ComPy 2.8 at YV.RR52, run with the ComPy 3 code, gives the archived draws exactly (maximum difference 0.0 over 8 x 5000 x 10 values). The additions to the sampler do not change a 2.x run. [test: `c01_legacy_identity.py`]

7. Rerun of the archive

All 287 stations with at least 100 compliance windows, with the wide-gate stage C results of the archive, covariances from the per-window spectra cache, and the tide-based response correction where the 2.9 audit has a leak factor (168 stations). Output: `~/Desktop/ComPy/14_inversion_overhaul/r01_archive/` (`tables/results_auto.csv`, one folder and one figure per station). Nothing earlier was overwritten. Total sampling time 5.5 hours of two stations at a time on six cores each; median 33 s per station; 141 of 172 tier-1 stations converged in the first round, 22 in the second, 9 used the third. [test]

7.1 Quality

	good	usable	poor
Tier 1 (172, for interpretation)	135	20	17
Tier 2 (115, nominal or doubtful gains, shallow water)	57	20	38

Reasons given for the 95 stations that are not "good" (a station can have several): misfit 63; no depth resolved 29; gain pulled more than 2 sigma 21 (17 of them nominal gains); not converged after three rounds 17; a Vp factor pulled 17; a resolved layer at a bound 6; density factor pulled 3. Of the 154 tier-1 stations that ComPy 2.8 called good, 127 are good, 19 usable and 8 poor now.

Median chi-square per datum is 1.08 (tier 1), against 0.38 in the 2.8 run: the stated errors now match the misfit.

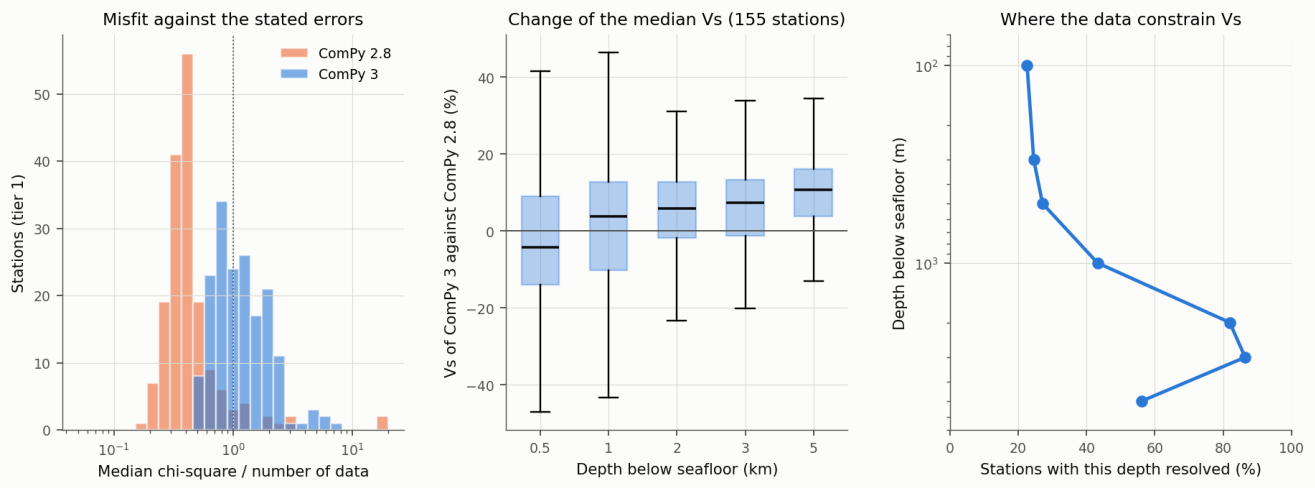


Figure 5. Tier 1. Left: median chi-square divided by the number of data, for the 2.8 run and the new run; 1 is what correct errors give. Middle: change of the median Vs between the two runs at five depths (box: quartiles, whiskers: range without outliers). Right: share of stations at which each depth counts as resolved.

7.2 The eight stations of the 2025 paper

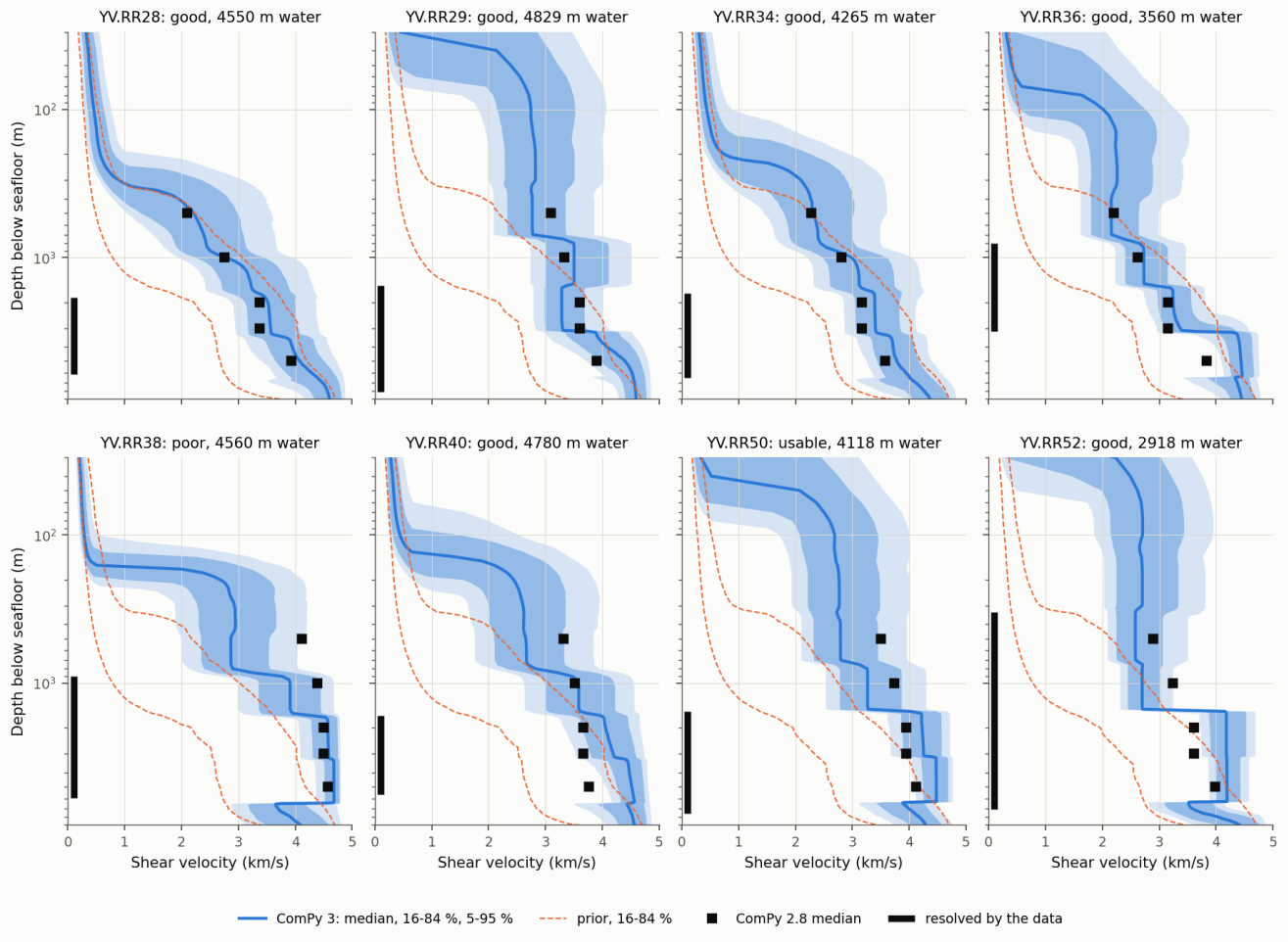


Figure 4. Vs against depth (logarithmic) at the YV stations. Blue: ComPy 3 posterior. Orange dashed: its prior. Black squares: medians of ComPy 2.8 at 0.5, 1, 2, 3 and 5 km. Black bar: depths resolved by the data. Where the bar is absent the blue band is the prior.

Median Vs in km/s with the 16 to 84 % interval; the 2.8 median in brackets; "u" marks an unresolved depth.

Site	Label	Soft sediment (m)	1 km	2 km	3 km	5 km
RR28	good	342 (242 to 555)	2.79 u (2.25 to 3.45) [2.76]	3.52 (3.11 to 3.99) [3.37]	3.56 (3.17 to 4.05) [3.37]	4.08 (3.65 to 4.51) [3.93]
RR29	good	32 (11 to 60)	3.51 u (3.00 to 4.11) [3.34]	3.29 (2.98 to 3.68) [3.61]	3.30 (3.00 to 3.72) [3.61]	4.40 (3.95 to 4.66) [3.91]
RR34	good	223 (154 to 332)	2.99 u (2.48 to 3.61) [2.81]	3.40 (3.04 to 3.83) [3.17]	3.41 (3.06 to 3.84) [3.17]	3.78 (3.43 to 4.17) [3.58]
RR36	good	73 (34 to 113)	2.73 (2.40 to 3.14) [2.62]	3.25 (2.97 to 3.56) [3.15]	3.37 (3.09 to 4.29) [3.15]	4.44 u (4.23 to 4.65) [3.84]
RR38	poor (misfit 26.6 for 6 data)	163 (130 to 201)	3.91 (3.35 to 4.44) [4.39]	4.58 (4.28 to 4.74) [4.50]	4.59 (4.30 to 4.74) [4.50]	4.68 (4.51 to 4.77) [4.57]
RR40	good	140 (98 to 190)	3.58 u (2.99 to 4.20) [3.52]	4.06 (3.64 to 4.47) [3.66]	4.19 (3.79 to 4.58) [3.66]	4.52 (4.26 to 4.72) [3.77]
RR50	usable (misfit 15.4 for 6 data)	42 (16 to 69)	3.27 u (2.78 to 3.86) [3.73]	4.23 (3.87 to 4.56) [3.95]	4.25 (3.89 to 4.57) [3.95]	4.48 (4.24 to 4.68) [4.12]
RR52	good	21 (7 to 40)	2.70 (2.44 to 3.02) [3.24]	4.18 (3.89 to 4.52) [3.61]	4.18 (3.90 to 4.52) [3.61]	4.19 (3.99 to 4.42) [3.98]

Three things to read from this table:

- **Vs of 4.1 to 4.6 km/s between 2 and 5 km at RR38, RR40, RR50 and RR52 is faster than gabbro** (3.7 to 3.9 km/s) and close to unaltered peridotite. It is not a general feature of the method: among the 76 deep tier-1

stations of the archive the median Vs at 3 km is 3.37 km/s (16 to 84 %: 3.11 to 3.67) and only 5 % exceed 4.0 km/s. So either these sites have thin crust or shallow ultramafic rock, which is plausible on the ultraslow Southwest Indian Ridge and near the Rodrigues triple junction, or the low-frequency compliance of the YV instruments is too low by a few per cent for a reason not yet identified. A slope error of 3 % moves Vs at these depths by about 6 %. This needs the judgement of people who know the sites and the instruments. [inferred]

- **About half of the rise against 2.8 is the Vp and density rule**, not the data: 2.8 sampled one Vp/Vs for all layers and found about 2.1, and its density followed that Vp. [inferred from the sensitivities of section 3.3]
- **The top kilometre is mostly unresolved** at these depths of water; values there are the prior.

The published profiles of the 2025 paper were not used anywhere in this work. How they compare with the table above is for the author to judge; ComPy 2.8 already differed from them (docs/INVERSION_2x.md).

7.3 Sediment against an independent map

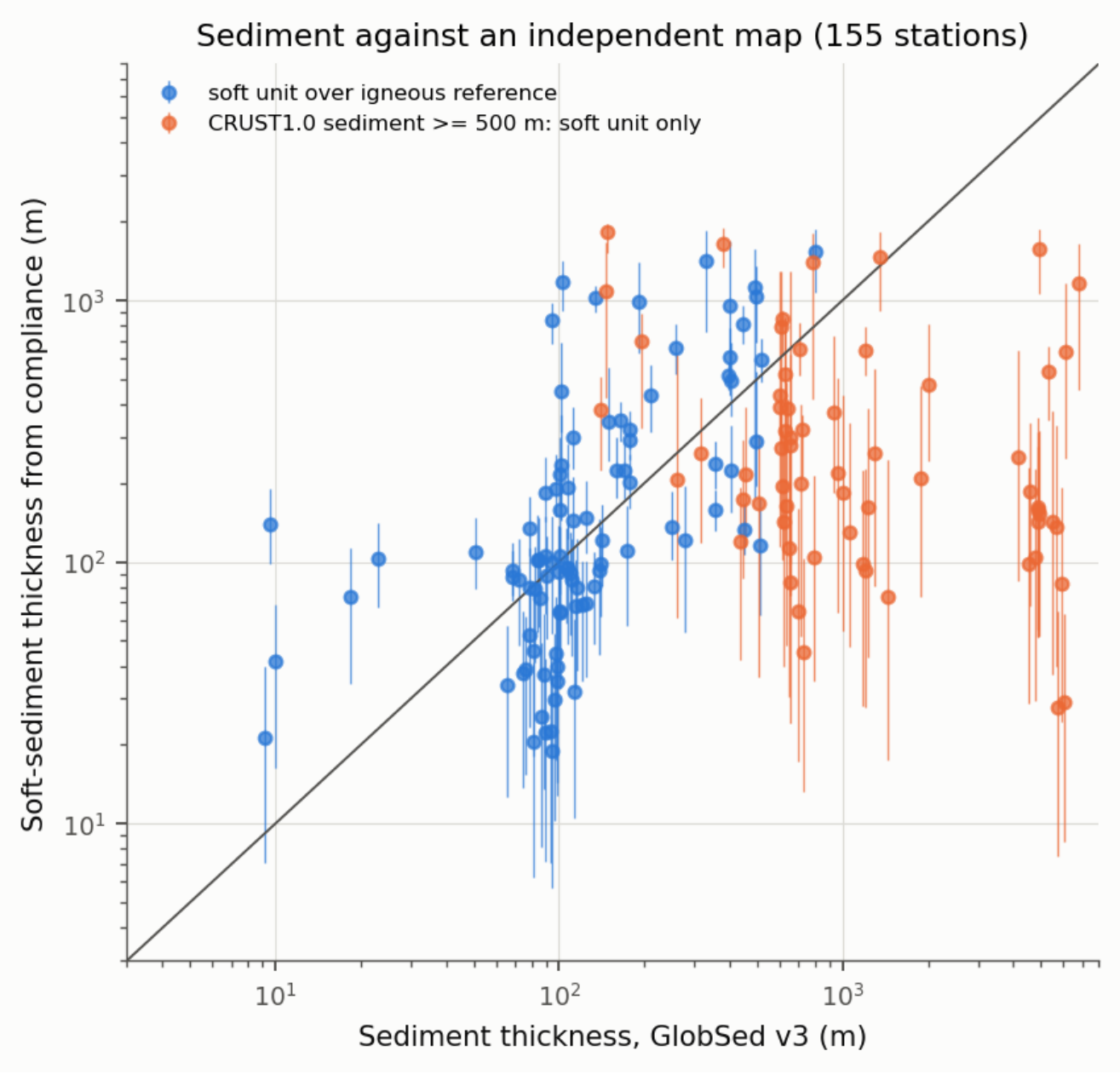


Figure 6. Thickness of the soft-sediment unit from compliance against total sediment thickness of GlobSed v3 (Straume et al. 2019) at the same position, tier-1 stations that are not poor. Blue: stations where the whole sediment column is the soft unit. Orange: stations where CRUST1.0 has at least 500 m of sediment, whose consolidated part is in the layered reference, so the soft unit is only its top. The line is equality.

GlobSed was not used by the inversion. For the 92 stations on thin sediment the rank correlation is 0.66 (p = 6e-13), the median ratio compliance/GlobSed is 1.09, and 79 % agree within a factor of 3. This supports the sediment term of the model, including its P modulus. The compliance thickness is a thickness of soft material: on young crust with 10 m of mapped sediment it gives 20 to 160 m, which is more likely rubble and porous lava than sediment. [test: `v01_globsed.py`, `g02_figures_results.py`]

No comparison with published compliance results of other groups at the same stations was made: Ruan et al. (2014), Bell et al. (2015) and Doran & Laske (2019) could not be obtained.

8. Sensitivity to the choices

Each choice was changed alone and the 43 benchmark stations were inverted again (`scripts/r02_batch.py`, output `r02_sensitivity/`). The table gives the change of the median Vs against the standard run, over the stations where the choice differs from the standard and neither run is "poor": median, 16 to 84 % range in brackets, and the largest change of any station. [test]

Choice changed	N	0.5 km	1 km	2 km	3 km	5 km
Prior width of ln Vs 0.5 instead of 0.3	17	+0.5 % (-7 to +5), max 9	-0.3 % (-4 to +3), max 6	-0.1 % (-2 to +2), max 5	+0.4 % (-2 to +2), max 5	+0.9 % (0 to +2), max 11
Correlation between layers 0.9 instead of 0.7	38	-0.4 % (-5 to +3), max 8	-0.5 % (-5 to +5), max 34	-0.1 % (-3 to +4), max 27	+0.9 % (-3 to +4), max 13	-1.5 % (-6 to 0), max 15
No response correction, slope allowance 4 %	34	+0.4 % (-1 to +2), max 6	+0.3 % (-1 to +2), max 5	-0.4 % (-3 to +1), max 6	-0.3 % (-3 to +1), max 11	+0.2 % (-7 to +2), max 48
Vp factor 15 % instead of 7 %	38	+0.1 % (-1 to +2), max 3	+0.2 % (-1 to +2), max 8	-0.1 % (-2 to +2), max 8	+0.3 % (-1 to +3), max 8	-0.7 % (-5 to +2), max 11
Band cut at 0.6 f_c instead of 0.8	35	+3.5 % (-3 to +24), max 44	+4.2 % (-1 to +21), max 71	-0.6 % (-22 to +12), max 47	+6.4 % (-3 to +20), max 49	-0.9 % (-6 to +4), max 21

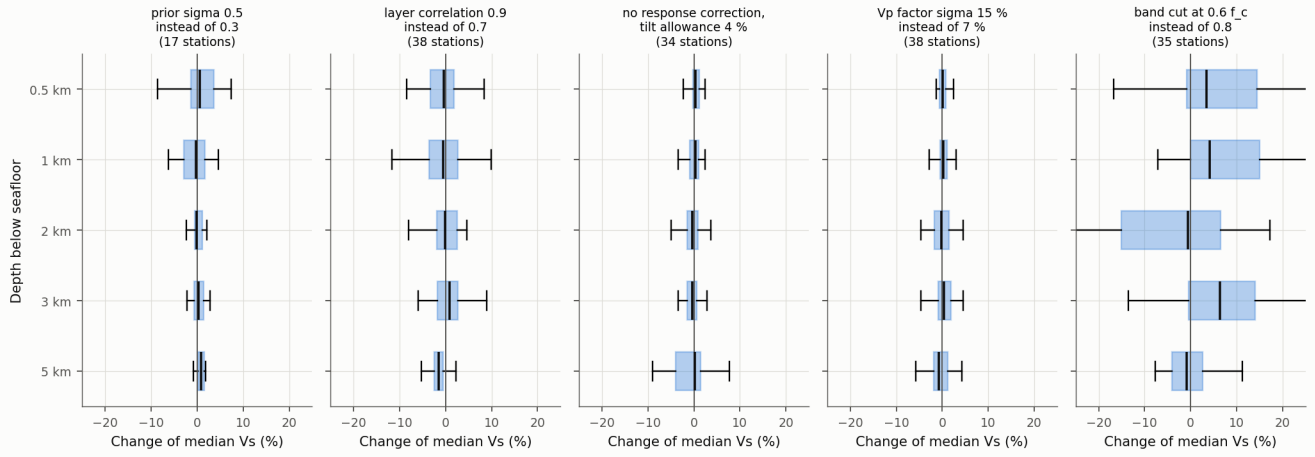


Figure 7. Change of the median Vs at five depths when one choice is changed, over the benchmark stations where the choice differs from the standard run (box: 25 to 75 %, whiskers: the range without outliers; the axis is cut at 25 %).

What the table shows:

- **The prior width of Vs and the Vp factor do not move the medians.** The medians change by less than 1 % over the stations and stay inside the 16 to 84 % interval of the standard run at every station and depth. They change the width of the intervals: 1.2 to 1.3 times wider down to 3 km with the wider Vs prior, 1.1 times wider from 2 km down with the wider Vp factor. The widths of the intervals carry these prior choices; the medians do not.
- **The correlation between layers** changes the median by at most 1.5 % over the stations and narrows the intervals to 0.7 to 0.9 of their width, but single stations change by up to 34 % at 1 km: a stronger correlation smooths a sharp step in the profile. The 24 benchmark stations with thick reference sediment already use a width of 0.5 and repeat the standard run exactly, which is why the first row has 17 stations.
- **The label "resolved" is relative to the prior.** It compares the width of the posterior with that of the prior, so a wider prior marks more depths as resolved. Of the 19 thin-sediment benchmark stations, 1 km is resolved at 7

with the standard width and at 15 with 0.5; 2 and 3 km are resolved at all 19 either way. With the stronger correlation between layers, 0.5 km is resolved at 31 of 43 stations instead of 22. Stations with thick reference sediment use the wider prior, so their shallow depths are marked resolved more easily than at the other stations. Medians and intervals do not depend on this; the label does.

- **The response correction** moves the median by less than 1 % in the median over stations and by up to 7 % at 5 km in the 16 to 84 % range. One station changes by 48 % at 5 km. Six quality labels change, four for the worse and two for the better.
- **The upper limit of the band is the choice that matters.** Cutting at 0.6 f_c removes three data per station (median) and changes Vs somewhere between 0.5 and 3 km by 20 % or more at 14 of the 35 stations; the new median lies outside the 16 to 84 % interval of the standard run at 14 to 37 % of the stations, depending on depth (32 % is what two independent estimates would give). The largest changes are at stations with thick sediment (7K.AN2 to AN6, 7D.M11B, ZU.NZ19), where the profile between 0.5 and 3 km alternates in sign with depth. Two stations have fewer than three frequencies left and cannot be inverted. Vs at 5 km changes little.

The last point has two readings, and this work does not decide between them. Either the bins between 0.6 and 0.8 f_c carry real information on the upper crust that the lower bins lack, and removing them returns those depths to the prior; or these bins are biased towards low compliance by the loss of pressure signal near f_c, beyond what the scatter trim removes, and bias the upper crust. Section 5 tests the inversion on synthetic data with the stated errors, so it does not test this. It is listed in sections 9 and 10.

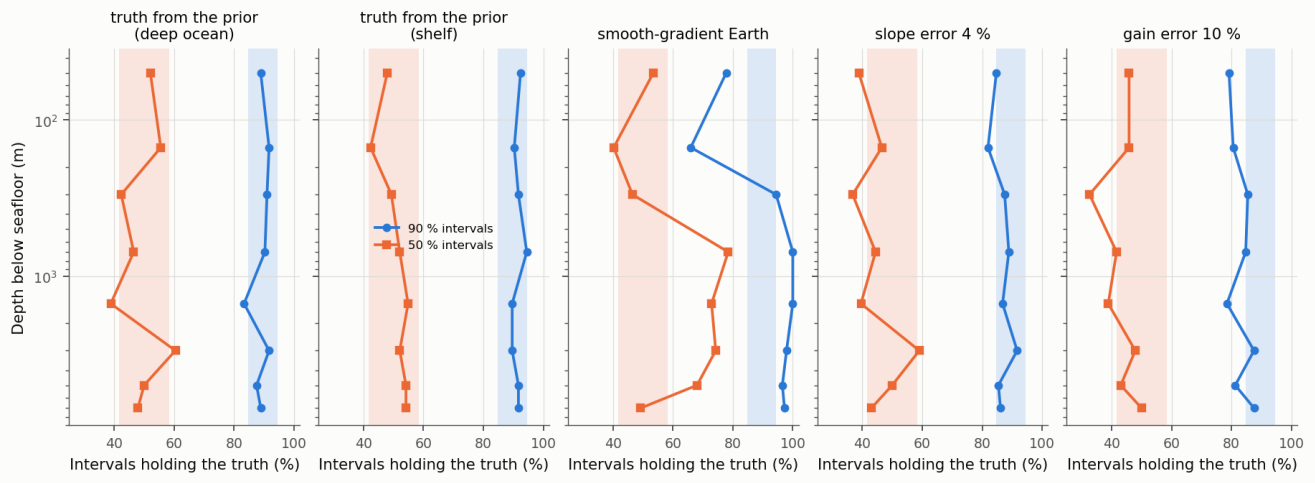


Figure 8. The calibration tests of section 5: share of synthetic data sets whose 50 % and 90 % intervals of Vs contain the true value, against depth, for five cases. Shaded: the range expected by chance for the number of data sets.

9. What failed, and what stays open

Item of the brief (section 3)	Status
Errors look too wide (chi-square/N 0.39)	Resolved: independent 1 % floor against 0.2 to 0.4 % correlated scatter. Now 1.08. [test]
Vs depends on the Vp and density rule	Explained (compliance measures stiffness) and handled by priors with sampled factors. Not removable: absolute Vs carries this assumption. A measured Vp profile can be given as the reference.
Vs depends on the method (minimum structure against Bayesian)	Not re-run. The minimum-structure code fixes Vp, density and the half-space and has no sediment P-modulus term; with the findings here the two answer different questions. Open.
Deep parameters press against their limits	Explained by the second-difference smoothing prior; gone with the proper prior, except where the data themselves ask for it (section 7.2). [test]
7K.AN3 at the 500 m/s bound; 7D.J09B does not fit	Both fit and converge with thick sediment in the reference and the band trim (chi-square 10.8 for 13 and 28.9 for 23). [test]
Fixed layer geometry narrows deep posteriors	The prior width is now explicit per layer and the Moho is free; resolution is reported per parameter. The layering itself is still fixed.

Item of the brief (section 3)	Status
Moho from CRUST1.0	Now a prior of 30 %. Posterior medians are 0.88 of CRUST1.0 (median, tier 1), but the Moho is weakly resolved (width ratio 0.5 to 0.9).
Sediment: Hamilton, density 2000, limit 1000 m; Ruan relation	Density from the Nafe-Drake curve, P modulus sampled, limit 2000 m. Ruan relation still not obtained and not used.
The paper's Metropolis sampler	Not changed. It stays as legacy code; its defects are listed in docs/INVERSION_2x.md.
H1 against H2	H1 kept, with evidence (section 4).
Pressure-response correction	Applied in the archive rerun where a leak factor exists (168 stations).
Band near the cut-off (7D.J44A)	Cap at 0.8 f_c plus the scatter trim; J44A is good (10.6 for 10).
Gain	Gaussian prior; pulled gains reported (21 stations).
Wide gates and their 15 dB cut; stage D without tilt cleaning where rates differ; pressure sign	Unchanged; not examined in this work. Open.
Saved curve stops at 40 mHz	Fixed in stage C.
The "resolved" label (new)	Open. It is a ratio of posterior to prior width and therefore depends on the prior width (section 8). An absolute criterion would not.
Upper limit of the band (new)	Open. Vs between 0.5 and 3 km depends on the bins between 0.6 and 0.8 f_c, most at thick-sediment stations (section 8). Whether those bins are biased was not tested.

Stations that remain poor and need a look at the data, not at the inversion: YV.RR38 (misfit, known pressure-channel change), 7D.J35A (BRUIT-FM gain 4.45; the inversion asks for a factor 0.83, Vp 25 % above the rock relation and Vs at 4.7 km/s), XO.LT01 (misfit 41 for 8 data in about 100 m of water), and the stations with nominal gains whose gain factor is pulled (17).

Not done: comparison with refraction velocities at the stations; a transdimensional or adaptive layering; the same audit for the compliance stages A to D, which earlier audits covered and this one only read.

10. Decisions for the author and W. Crawford

1. **Publish ComPy 3.0.0?** The branch changes the default Vs results. It is pushed as a pull request and not merged.
2. **The slope allowance** (2 % with a response correction, 4 % without) and **the prior widths** (ln Vs 0.3, Vp 7 %, density 3 %, Moho 30 %). These are judgements; section 8 shows what each does.
3. **The fast lower crust at RR38, RR40, RR50 and RR52**: geology or a low-frequency bias of a few per cent? Refraction or gravity constraints at these sites would settle it.
4. **The upper limit of the band** (0.8 f_c). Section 8 shows that shallow Vs depends on it. A test on real data would be to compare compliance of the same site from two water depths or from a pressure gauge with a known noise floor.
5. **The criterion for "resolved"**: a ratio to the prior width (now), or an absolute posterior width? Section 8 shows that the ratio favours stations with a wide prior.
6. **A Vp profile per site.** Where refraction Vp exists, giving it as the reference removes most of the Vp/Vs uncertainty (Crawford 1994, 2004 did this). Should the pipeline look for one?
7. **The corrections to the 2025 paper** now known: the sign and size of the gravity correction in the text (section 3.1), and the items of docs/INVERSION_2x.md. Whether to publish a correction.
8. **Sources to check by hand**: Crawford et al. (1991, 1998), Ruan et al. (2014), Bell et al. (2015), Brocher (2005), Hamilton (1976, 1979), Carlson & Raskin (1984). The relations taken through secondary sources are marked in the code.
9. **The 2.x results in the README and in docs/results/** (XE, YV) were made with ComPy 2.4 to 2.8. They are marked as superseded; rerunning those networks through the full pipeline with 3.0 is a separate job.

11. Files

- Code: `compy/inversion/auto.py`, `errors.py`, `cli.py`, `compy/plotting/auto.py`; changes in `compy/inversion/dram.py`, `pipeline/c_compliance.py`, `e_invert.py`, `f_figures.py`, `g_report.py`; tests `tests/test_inversion_auto.py`.
- Scripts of this report: `scripts/` (`e01` to `e04` tests, `c01` legacy identity, `r01` to `r03` archive runs, `v01` GlobSed, `g01`, `g02` figures, `archive.py` inputs).
- Outputs: `~/Desktop/ComPy/14_inversion_overhaul/`.
- Decisions: `~/Desktop/ComPy/DECISIONS.md`, D49 to D57.